F78AA - Actuarial and Financial Mathematics A

Tutorial 6 – Bonds and Tax

Question 1 A government has just issued a fixed interest security (or bond) of nominal amount £100 pays a coupon of 6% p.a. at the end of each of the next ten years and is redeemed at par at the end of the ten years.

1. Determine the price of the security which gives a redemption yield of 6% p.a. effective to a purchaser who pays no tax. Explain the answer.
2. Calculate the price of the security that gives a redemption yield to the same purchaser of 8% p.a. effective.

Question 2 Suppose that in the above example the coupon is £6 p.a. payable half-yearly.

1. Calculate the price of the security that gives a yield to the same purchaser of: (a) 6% p.a. effective, and (b) 8% p.a. effective.
2. Explain why your result in (a) is not £100.

Question 3 A fixed interest security of nominal amount £100 000 will be repaid at 110% after 15 years. The security bears a coupon of 9% p.a. payable annually in arrears. On the issue date, the security is purchased for £80 000 by an investor who is liable to income tax at 40% and to capital gains tax at 30%.

1. Calculate the investor's net (meaning "after tax") effective redemption yield p.a. for the transaction.
2. The investor wishes to secure an after-tax redemption yield of 9% p.a. Determine how much the investor should pay for the security.

Question 4 Consider a 20-year bond, bearing a half-yearly coupon of 8% per annum and redeemable at 120%. It is exactly six months until the next coupon. Calculate the price of the bond to give a redemption yield of 9% per annum effective after tax to an investor who pays:

1. no taxes.
2. income tax at 20% but no capital gains tax (CGT).
3. income tax at 20% and CGT at 30%.

Question 5 On 1 January 2003, a corporation issues a number of bonds, each of nominal amount £100, to be redeemed at 98% on 1 January 2028. The coupon is 7% per annum, payable half-yearly on each 1 January and 1 July.

1. The bond is priced at issue to give a redemption yield of 8.16% per annum to a non-tax-paying investor. Calculate the issue price per £100 nominal.
2. Anna, who pays income tax at 25% and CGT at 40%, bought some of the bonds on the issue date. On 1 July 2013, just after the coupon then due was paid, she sold the bonds to Bob, who pays both income tax and CGT at 40%. Bob paid a price that gave him a redemption yield after taxes of 6% p.a. effective. Calculate the price that Bob paid per £100 nominal.
3. Calculate the yield realised by Anna on her completed transaction.

Question 6 Investors often allow for risk by requiring a higher yield to lend money to a borrower who is thought to be at higher risk of not repaying some or all of the interest and capital. Explain the consequences for companies or governments who are trying to raise money in the bond markets.

Question 7 An investor buys a 20-year bond, to be redeemed at 110% and bearing semi-annual coupons at rate 6% per annum, to obtain a redemption yield of 5% p.a. effective. The investor pays income tax at 25% but no CGT.

1. Calculate the price paid by the investor.

2. Just after the 20th coupon payment, the investor's rate of income tax drops to 20%. If she holds the bond to redemption, calculate her realised yield on the whole transaction.

Question 8 A fixed interest stock with nominal value of £100 was issued on 1 January 1995. The coupon rate was 4% p.a. payable half-yearly in arrears for the first 10 years and then 3% p.a. payable half-yearly in arrears for the next 10 years. Redemption was at par after 20 years. On 1 July 1999, just after the July dividend was paid, the stock was quoted at a price which would have given a tax-free investor a redemption yield of 6% p.a. The stock was bought on that date by an investor who was subject to income tax at 40% and to capital gains tax at 20%.

1. The investor held the bond until the redemption date. Calculate his after tax redemption yield.

2. On 1 August 2000 the stock was priced to give an after tax redemption yield of 5% p.a. effective to the taxed investor described above. Calculate the price of the stock on 1 August 2000. (Note: be careful with the dates.)

Solutions: Q1.1. £100; Q2.1. £86.57984; Q2.1(a). £100.65278; Q2.1(b). £87.37017; Q3.1. 0.0772; Q3.2. £70.47126; Q4.1. £96.04715; Q4.2. £81.12001; Q4.3. £78.92111; Q5.1. £88.97748; Q5.2. £79.42754; Q5.3. 5.12% p.a.; Q7.1. £98.23044; Q7.2. 5.001% p.a.; Q8.1. 3.999% p.a.; Q8.2. £67.17545.